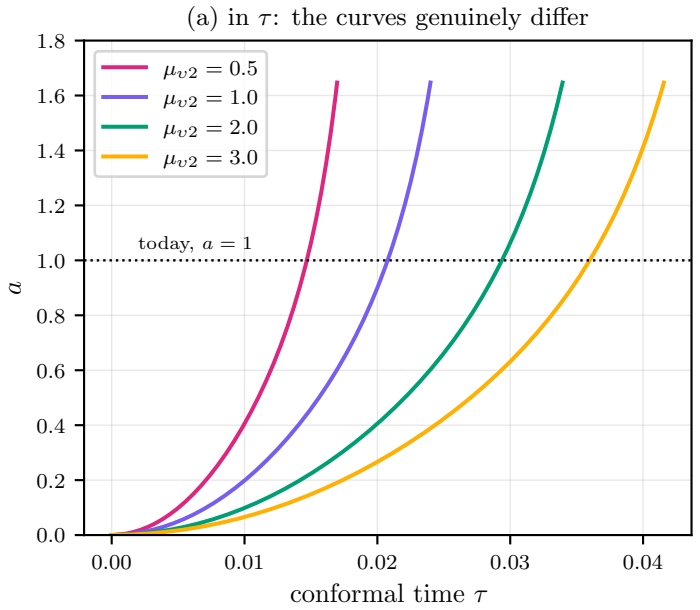
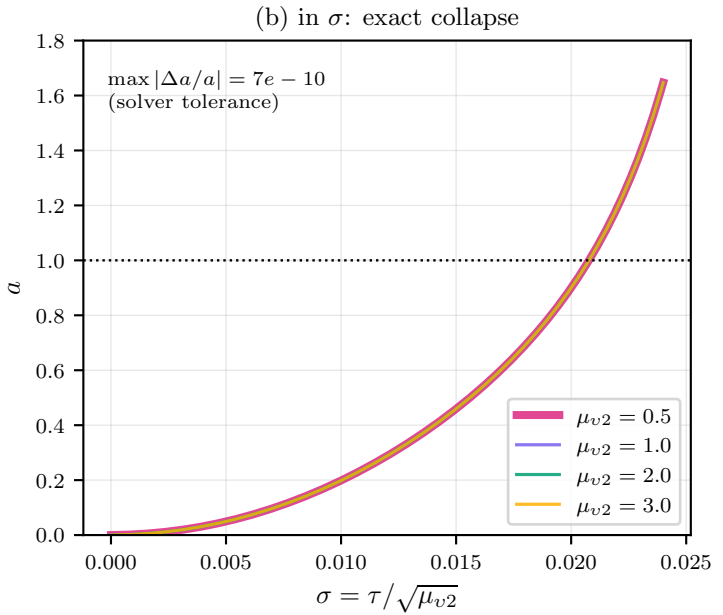
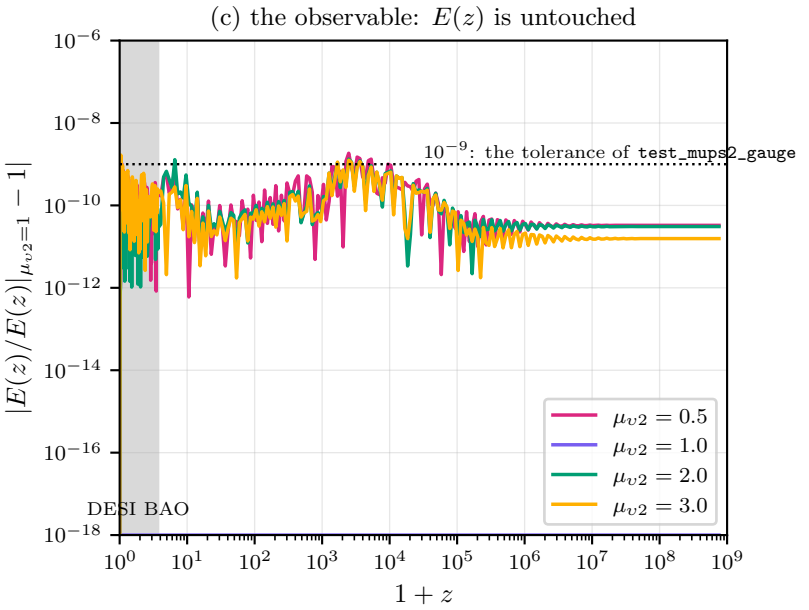
Why μ_{v2} does not affect distances: it rescales the field's clock, $\tau = \sqrt{\mu_{v2}}\,\sigma$, and nothing else







In the field equations μ_{v2} multiplies every term quadratic in τ-derivatives but none of the matter/radiation/Λ sources, so it is absorbed entirely by $\tau = \sqrt{\mu_{v2}}\sigma$; the series initial conditions scale the same way ($\tau_{\rm ini} \propto \sqrt{\mu_{v2}}$, term $n \propto \tau^n \mu_{v2}^{-n/2}$).
The trajectory $a(\sigma)$ is therefore *identical* for every μ_{v2}, and $H \propto a^{-2}\,{\rm d}a/{\rm d}\tau$ picks up one common factor $\mu_{v2}^{-1/2}$ at every epoch. $E = H/H(a{=}1)$ is a ratio, so it cancels identically — for any H_0, sampled or not.
μ_{v2} is a gauge, like a_0 and h. It becomes physical only where a scale that does *not* co-rescale with τ enters — e.g. a comoving k at perturbation level.